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Practical Aspects of Impairment Rating and Disability Determination

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SUMMARY PEARLS

- Physiatrists are uniquely qualified among medical specialists to ascribe impairment ratings and perform disability evaluations.
- A biopsychosocial model is preferred for conceptualizing disability.
- The American Medical Association's *Guides to the Evaluation of Permanent Impairment*, Sixth Edition, is widely used to understand qualitative and quantitative impairments and ascribe overall impairment ratings.
- Independent medical examinations (IMEs) are typically one-time evaluations performed by a physician who is not directly involved in treating a claimant.
- IME examinations can provide information that can be subsequently utilized in litigation claims and/or other legal proceedings, among other applications.
- Physiatrists can participate in various aspects of medicolegal work, including expert witness testimony, IME, and life care planning.
- Physiatrists performing medicolegal work should be aware of the liabilities associated with such practice.

Introduction

The field of physical medicine and rehabilitation is primarily devoted to the clinical evaluation and treatment of disabling sequelae of illness, injury, infirmity, or deformity across the lifespan. As such, physiatrists can expect to periodically be called on to make formal disability assessments of their patients. Furthermore, the face of medicine continues to evolve, brought about by a combination of advances in medical and surgical technology and an increase in health awareness that helps people survive and live longer even after catastrophic trauma or illnesses. According to the latest data from the Centers for Disease Control and Prevention's Disability and Health Data System, as of 2021 approximately 66 million (27.2%) Americans aged 18 years or older are currently living with one or more disabilities.¹⁴ The prevalence of disability is even higher among American military veterans (>30%) compared with the overall adult population, and highest (>43%) among Americans over age 65 years,¹⁴ with these numbers expected to continually increase during the next decade. Further, it is estimated that 37 million baby boomers will have more than one chronic condition by 2030, contributing to a surge disability prevalence within the United States.⁹ This impending shift in demographics and increasing prevalence of disability afford additional opportunities for physiatrists to adapt their practice to optimally address the health care needs of patients through acquisition of a greater understanding of the

conceptual foundation and terminology of disablement, and application of the same to the practices of impairment rating, disability determination, and life care planning.

This chapter includes a brief historical perspective of disability evaluation, a conceptual foundation for understanding the current models of disablement, a descriptive overview of key features and nuances of the various US disability systems, a working vocabulary and practical application of tools available and procedures to follow for proper and valid performance of impairment ratings and disability determinations (including use of the most relevant physician guides), an introduction to extensions of impairment ratings for life care planning, and a heightened awareness of the medicolegal framework, pitfalls, and ramifications of such undertakings.

Terminology and Conceptualization of Disablement

During much of the 19th and 20th centuries, disability was conceptualized in terms of a medical model of disease, whereby causation was viewed as directly linked to some underlying pathologic condition potentially identifiable at a histologic and physiologic level and arising out of illness or trauma. Because the diagnosis and treatment of disease was the purview of physicians, diagnosis and treatment of the resulting disability also fell within their expected domain of expertise.^{32,51} With the advent of the stethoscope,

microscope, x-ray, and other technologies, the physician was armed with instruments of precision and objectivity with which to rate severity of impairment (at an organ system level) and hence objectify the associated disability. The model worked well for disease entities that were relatively devoid of diagnostic ambiguity and where the disease was well understood, as well as for treatment goals and expected therapeutic endpoints.^{59,106} Today, the medical model still serves as the fundamental basis for Social Security disability determinations and for physician rating schedules that remain largely anatomically and diagnostically based.

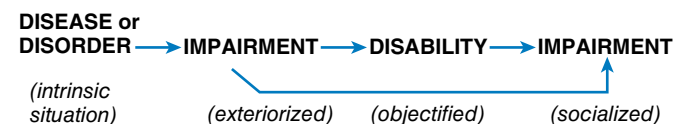
A social model of disability grew out of the advocacy movement of the 1970s and 1980s, making society accountable for disability in the face of disease through its failure to accommodate the special needs of individuals with disability in terms of environmental access, availability of adaptive equipment, discrimination, prejudicial thinking, and other attitudinal barriers at the time.⁵¹ A better documentation and understanding of the role of social barriers to functioning has helped foster strategies and tactics to neutralize the same and to enable and empower the members of our society with disability.²⁷

A biopsychosocial model of disability^{24,25} is now the preferred model and has gained wide acceptance when disability is conceptualized. The biologic component refers to the physical or mental aspects, or both, of an individual with a given health condition; the psychological component recognizes personal beliefs, coping strategies, emotional, and other psychological factors that may affect functioning; and the social component recognizes contextual, infrastructural, and other environmental factors that may also affect functioning in any given case.^{105,106}

From Classification of Causes of Death to International Classification of Functioning, Disability, and Health

The World Health Organization (WHO) has created an international classification system of diseases and disablement whose origins can be traced directly to the publication of Bertillon's *Classification of Causes of Death*¹² and, subsequently, the *International Statistical Classification of Diseases, Injuries, and Causes of Death*.¹⁰⁹ By 1948, the WHO began leading this effort and ultimately created the *International Classification of Impairments, Disabilities, and Handicaps* (ICIDH),¹⁰⁷ with the conceptualized framework reproduced in Fig. 6.1.

This system was an attempt to relate the pathologic features of a specific disease or trauma to the resulting impairment (physiologic consequences in terms of signs and symptoms of dysfunction at an organ system level), disability (functional consequences of impairment in terms of abilities lost in one's personal sphere), and handicap (social and societal consequences, freedoms lost in terms of role fulfillment). The relationship lends itself to being



• **Fig. 6.1** The World Health Organization International Classification of Impairments, Disabilities, and Handicaps. (Modified from World Health Organization. *International Classification of Impairments, Disabilities, and Handicaps: A Manual of Classification Relating to the Consequences of Disease*. WHO; 1976.)

depicted in a simplified linear manner, which implies unidirectional nature and causation, and fails to adequately account for confounders of a personal and environmental nature. More recently, the ICIDH has been replaced by the *International Classification of Functioning, Disability, and Health* (ICF),¹⁰⁸ with the conceptualized framework reproduced in Fig. 6.2.

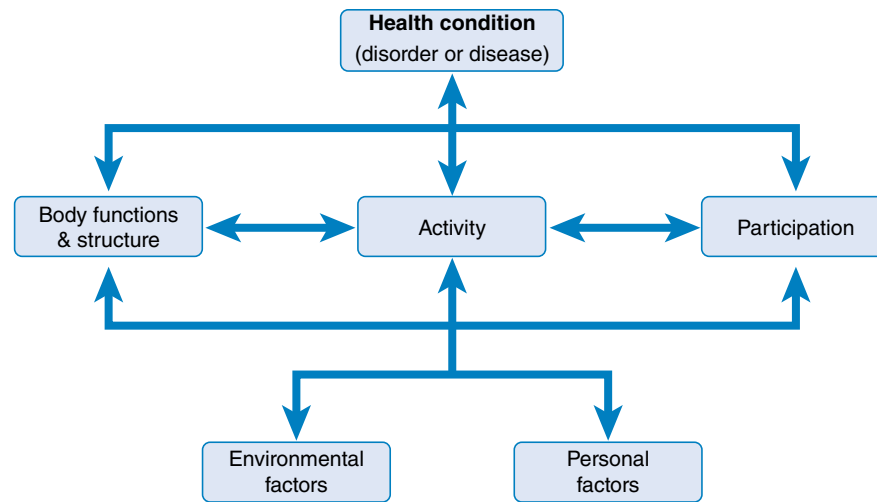
The ICF system more aptly displays the interactive (i.e., non-linear) relationships between the impairment that an individual afflicted with a particular health condition might face, the potential functional consequences of impairment with respect to that individual's personal and social sphere, and the contextual factors that may mitigate or amplify these consequences, including environmental factors, personal experiences, and choices.¹⁰⁸ By taking environmental and personal factors into account, the ICF has adopted the biopsychosocial model of disablement discussed earlier.^{24,25,105,106} Given this significant strength, use of the ICF has grown around the world over the past 20 years throughout complementary sectors, including clinical medicine, education, and governmental policy.⁴⁴ The components of disablement according to the ICF classification system include:

- Body functions and body structures: physiologic functions and body parts, respectively
- Activity: execution of a task or action by an individual (typically functioning within a person's own personal sphere)
- Participation: involvement in a life situation (typically within one's social sphere)
- Impairments: problems in body function or structure, such as a significant deviation or loss
- Activity limitations: difficulties that an individual may have in executing activities
- Participation restrictions: problems that an individual may experience in involvement in life situations

The ICF offers a conceptual platform for identifying the disabling consequences of impairment to an individual with a health condition (disorder or disease), whereby the modifying influence (in a positive or negative sense) of environmental and personal factors can also be recognized and more properly accounted for.^{44,108}

Americans With Disabilities Act and Implications

With passage of the Americans With Disabilities Act (ADA) in 1990,⁴ Americans with disabilities were guaranteed equal rights to employment opportunities, transportation, and public access. The ADA has its own key terminology and defines disability as "a physical or mental impairment that substantially limits one or more of the major life activities of such individual, a record of such impairment or being regarded as having such an impairment."¹ Although it is broad and somewhat imprecise, this definition is narrowed under Title I of the ADA (Employment) to recognize employment as a major life activity and views disability within the context of performance of the essential functions of an employment position with or without reasonable accommodation. Reasonable accommodation can include structural modifications of the worksite to improve accessibility, availability of modified duty options, and acquisition of adaptive equipment or devices to enable an otherwise qualified worker with a disability to perform the essential functions of the job.³ Accommodations exempted under ADA include those that pose "undue hardship" to the employer in terms of cost or feasibility of implementation



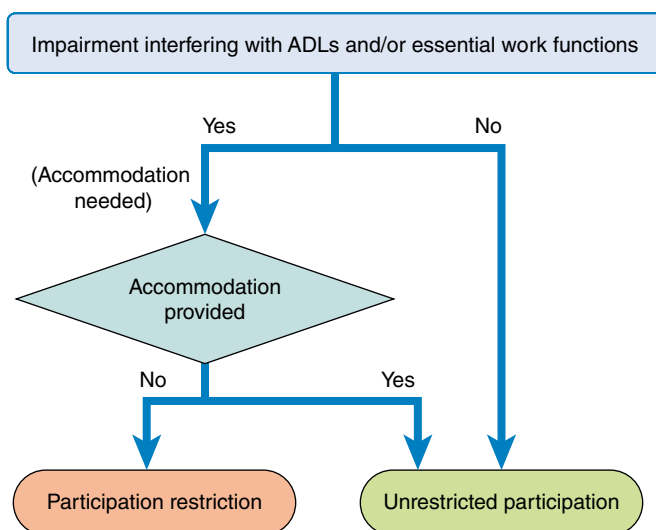
• **Fig. 6.2** International Classification of Functioning, Disability, and Health model. (From World Health Organization. How to Use the ICF: A Practical Manual for Using the International Classification of Functioning, Disability and Health (ICF). Exposure Draft for Comment. WHO; 2013.)

or those that would pose a “direct threat” to the health and safety of the individual with a disability and/or coworkers.¹

As can be seen from the preceding ICF discussion, accommodation under the ADA is a fundamental social and environmental modifier mandated by statutes to mitigate the disabling consequences of impairment in the workplace in terms of accessibility with respect to activity limitations and participation restrictions. The ADA perspective can be further conceptually integrated with this broader ICF terminology and definitions by viewing impairment, reasonable accommodation, activity limitations, and participation restrictions in relation to a major life activity, such as in a workplace (Fig. 6.3). If the required accommodation is provided, then no disability in terms of activity limitations needs to exist. If accommodation is not judged reasonable or otherwise cannot be provided, then disability in terms of activity limitations exists. If the impairment potentially interferes with the ability to safely and efficiently

perform the essential functions required of the job, and specific accommodation is required and can be provided, then no disability in terms of participation restrictions needs to exist relative to that job. If the accommodation required is not reasonable (poses undue hardship or direct threat) and is thereby denied, then disability in terms of participation restrictions exists relative to that job. Although participation restrictions in the workplace can arise out of impairment, they may not be determined solely (or primarily) by the impairment itself because other contextual factors might be the primary determinant of disability in any particular case. Consider, for example, the vastly differing work-related consequences of a partial foot (e.g., first digital ray transmetatarsal) amputation to a ballerina who must perform *en pointe* and to a construction worker who habitually wears steel-toed work boots on the job. The former amputee might face career-ending changes, whereas the construction worker could arguably experience minimal participation restrictions for the same level of impairment by using minor footwear modifications.

Although the ADA helps ensure that individuals with disabilities cannot be discriminated against in the workplace and that reasonable accommodation in terms of access must be provided to avoid such discrimination, the workers’ compensation system is not presently directly accountable to this legislative mandate. However, the physician examiner seeking to return an injured worker safely to the workplace is well served to review the worker’s formal job description and to tailor the ensuing therapy treatment plan to the specific material handling and activity requirements listed according to the essential functions therein. In addition, return-to-work decisions can be driven by the claimant’s observed performance in therapy according to these essential functions, thereby releasing the claimant to return to work when the worker’s physical performance meets or exceeds these requirements, or by recommending restricted duty and suggesting accommodations when they do not.⁶⁴ The employer, however, is ultimately responsible for determining reasonable accommodation. It is not the responsibility of the disability-evaluating physician to determine the essential functions of a job, to devise accommodation, or to determine reasonableness of any accommodation proposed by the employer.³⁵



• **Fig. 6.3** Relationship of impairment, accommodation, and participation restrictions in the workplace. ADLs, Activities of daily living.

Relating Impairment to Disability and Compensation Formulas

Although the major current disability systems of the United States and elsewhere exhibit significant differences from one another (discussed later), they all share a common mandate: to compensate individuals financially for losses because of their qualifying disablement. The challenge becomes one of providing fair and equitable compensation for losses that typically can be expected to occur in three major domains: work disability, nonwork disability, and quality of life (Fig. 6.4).^{33,60}

An impairment rating provides an objective measure of severity of disability in terms of organ system disease and associated loss of structure and function. As such, it is a keystone to any disability determination but not the sole or necessarily adequate determinant of losses to the affected individual. Metrics also exist to calculate losses to the impaired individual in terms of work disability (loss of earnings or earning capacity) and are commonly applied when actuarial analysis takes place. Less obvious, and perhaps less accessible, are metrics to assess avocational (e.g., non-work) disability, such as losses in ability to pursue hobbies and recreational activities, and overall quality of life,⁴⁷ including losses in terms of medical burden of care^{23,48} and overall life satisfaction. These more complex domains of disability are typically overlooked for the sake of expediency, simplicity, and practicality and are often gratuitously (rather than empirically) accounted for. A procedural shortcut typically takes place whereby the impairment rating becomes a surrogate for the disability rating according to a predetermined formula whereby the impairment percentage is multiplied times the number of weeks arbitrarily established as the “worth” of the whole person times a percentage (usually two-thirds to three-quarters) of the average weekly wage (up to a cap) to generate a lump-sum payout.⁵⁹

In summary, impairment rating is important to the categorization of disablement for the following reasons⁵⁹:

- It serves as a standard reference point in terms of linking a specific diagnosis to an associated percentage of physical and functional loss in compensable injury claims.
- It enables the impaired individual to exit the system of temporary disablement (e.g., temporary total or permanent partial

disablement under workers’ compensation) at maximum medical improvement (MMI).

- It provides a diagnosis-based classification of severity to segue to alternative systems management of long-term disablement.

Major US Disability Systems Compared With Attention to the Role of the Examining Physician

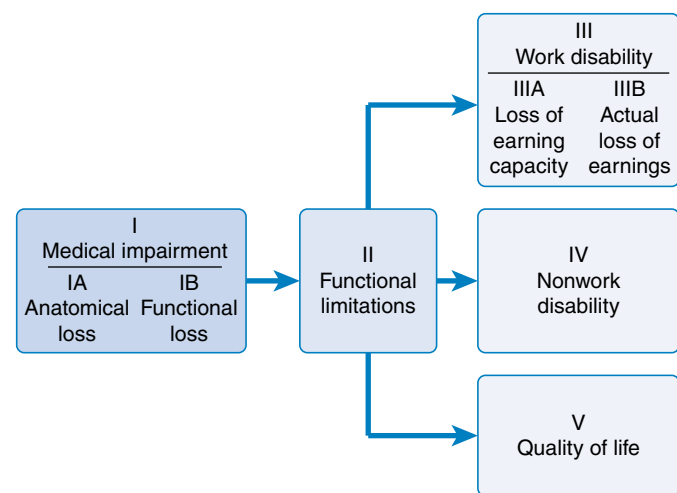
The physician who deals with claims of impairment and disability and issues regarding return to work or stay at work should become familiar with the multiple US compensation systems, which exist both at the federal and state levels, to provide economic and other benefits to claimants experiencing work incapacity because of injury, illness, or aging (reviewed in Katz³⁸). Chief among these are the Social Security Administration (SSA), federal and state workers’ compensation systems, the Veterans Benefits Administration (VBA), and personal injury claims. A physician working within the jurisdictional boundaries of any of these systems plays a critical role and is expected to diagnose the problem, opine as to the causation of an injury or illness (particularly in a workers’ compensation setting), provide care, and assess the work fitness of the injured claimant for “modified duty” or return to “full duty” during recovery. Therefore the physician must also become familiar with the definitions, eligibility requirements, and rating criteria of each system and must follow the system-specific rules and procedures prescribed in each particular case.

Social Security Disability Insurance and Supplemental Security Income

The SSA is the largest US disability system^{58,103} providing assistance to persons who might otherwise qualify as disabled and can meet the specific requirements of either of two respective pathways: Social Security Disability Insurance (SSDI), under Title II of the Social Security Act,⁹⁸ and Supplemental Security Income (SSI), under Title XVI of the Social Security Act.¹⁰¹

The first pathway, SSDI, provides benefits under Title II to those individuals who have worked in a qualified job for a requisite period, paid into the Social Security system, and subsequently become disabled before age 65 years.^{58,98,103} SSDI is funded by payroll deductions that, in combination with deductions for old-age insurance, comprise the Federal Insurance Contribution Act tax, with matching contributions from the employer. Eligibility for SSDI is detailed in *Blue Book: Disability Evaluation Under Social Security*,¹⁰² which requires that a “medically determinable impairment” be established whose resulting incapacitation is so severe that it prevents the affected individual from engaging “in any substantial gainful activity ... by reason of any medically determinable physical or medical impairment(s) that can be expected to result in death or that has lasted or can be expected to last for a continuous period of not less than 12 months.”¹⁰² SSDI also requires that the individual has worked in a qualified job for at least 5 of the 10 years before onset of disability. SSDI benefits are provided to those considered totally incapacitated and extend to their surviving spouse and children. Benefits are paid as a monthly stipend, and beneficiaries may receive payments until age 65 years, after which they are eligible for Social Security retirement benefits.^{58,103}

The second pathway, SSI, provides benefits under Title XVI to “individuals (including children under age 18) who have limited



• **Fig. 6.4** Disabling consequences of an injury or disease. (Modified from Institute of Medicine. *A 21st Century System for Evaluating Veterans for Disability Benefits*. 2007;117. <https://doi.org/10.17226/11885>.)

income and resources” and one or more qualifying “medically determinable impairment,” with eligibility determined according to a means test and not the SSDI-required work history.^{58,97,101-103} SSI operates as a federal-state partnership that is jointly funded through general revenue (federal and state income tax).^{38,58,102}

When an applicant submits an SSA application (either SSDI or SSI) and nonmedical eligibility has been established, the application is forwarded to the respective state’s Disability Determination Service (DDS) for further medical review.¹⁰² The SSA has a defined set of medical criteria, the “listing of impairments,” arranged by body system for both adults⁹⁹ and children.¹⁰⁰ Each listing typically contains a diagnosis and some clinical markers of severity. Meeting listed medical criteria will result in an automatic award of benefits. If listed medical criteria are not met, the applicant can appeal on the basis of “residual functional capacity.”⁵⁸

Physiatrists who assist applicants when filing for SSDI or SSI disability should be familiar with listing of impairments,^{99,100} the defined process to evaluate the medical evidence,¹⁰² and the appeals process.⁹⁶ They may be asked to provide the DDS evaluating team with a statement about the patient’s ability to do work-related activities, backed by objective evidence. They may also be asked to comment on an applicant’s physical and psychological capacities and limitations and, if the condition in question does not meet or equal the listings, to assist the DDS evaluating team in estimating the claimant’s “residual functional capacities.”⁵⁸

Federal Workers’ Compensation Systems

In the United States, workers’ compensation laws originated at both the federal and state levels at the beginning of the 20th century and have evolved over the past 100 years into the current programs to provide no-fault–based compulsory broad coverage of injuries and illnesses arising out of and in the course of employment.³⁸

The major federal workers’ compensation programs are administered by the US Department of Labor’s Office of Workers’ Compensation Programs (OWCP).⁸⁷ The largest program is the Federal Employees’ Compensation Program,⁸³ which covers over 2.9 million federal employees under the Federal Employees’ Compensation Act,⁸⁹ along with a number of other worker groups adopted by Congress through various acts of expansion of the federal authority.^{42,90} In addition, the Black Lung Program,⁸⁰ Energy Workers Program,⁸² and Longshore Program⁸⁵ are other federally mandated compensation acts administered by OWCP. The US Department of Labor maintains exclusive jurisdiction over these programs, which are subject to judicial review. These four major federal workers’ compensation programs provide wage replacement benefits, medical treatment, vocational rehabilitation, and other benefits to injured workers (or their dependents) who experience work-related injury or occupational disease.^{38,87}

Other federally mandated workers’ compensation programs include the Federal Employers Liability Act (commonly known as the Railroad Worker Act),⁹⁴ the Jones Act (officially titled the Merchant Marine Act of 1920),⁹⁵ and the Defense Base Act (DBA).^{81,93} The DBA provides workers’ compensation protection to civilian employees working outside the United States on behalf of the US government or any other “American employer providing welfare or similar services for the benefit of the Armed Forces pursuant to appropriate authorization by the Secretary of Defense; irrespective of the place where the injury or death occurs.”⁸¹

State Workers’ Compensation Systems

At the individual US state level, each state has enacted a workers’ compensation law that protects the injured worker for injuries arising out of and in the course of employment. A complete list of contact information and state government websites for all US state workers’ compensation officials is maintained by the US Department of Labor’s OWCP.⁸⁶ The individual workers’ compensation laws within each state have their own nuances; however, the following fundamental features are common to all these statutory schemes⁶³:

- Compulsory insurance is required for employers with very few exceptions.
- A no-fault system exists for injuries arising out of and in the course of employment.
- Expedited benefits are included for medical and rehabilitation treatment and for wage replacement.
- Benefits also include compensation for permanent partial and permanent total disability.
- The law ensures access to an exclusive source of remedy for employees for work-related injuries/illnesses with few exceptions.
- The injured worker retains the right to sue any third party liable for injury.
- Disputes are usually resolved through administrative law adjudication with less rigorous rules of procedure and evidence.
- Administration is through a state agency designated to oversee workers’ compensation law.

Workers’ compensation benefits include survivor benefits (in the event of death), coverage of medical and rehabilitative expenses, and wage-loss benefits (which typically provide two-thirds of weekly wages up to a cap) during the healing period of temporary disablement. Disability under workers’ compensation can be determined to be temporary or permanent, partial or total. Many states have also enacted a second injury fund, whereby employment of individuals with preexisting work-related disabilities is encouraged. Such funds can reduce employer risk (e.g., cost of insurance) for additional compensation and medical expenses in such cases.⁴⁹

The physician examiner can play a variety of roles in the work-related injuries associated with workers’ compensation claims.^{16,36,48,60,62,102} There are four possible phases of involvement, including (1) initial evaluation and treatment of injury either as an approved and designated attending physician or as an authorized consultant; (2) overseeing rehabilitation, including return-to-work or stay-at-work issues; (3) determination of any residual impairment (permanent) or disability (work restrictions); and (4) estimating long-term care needs in catastrophic injuries (e.g., limb amputations, spinal cord injuries, major multiple trauma), including participation in life care planning. The physiatrist can be particularly successful in meeting the needs of the workers’ compensation system by maintaining functionality as a key focus when setting treatment goals and monitoring progress toward the same, by enabling a return-to-work treatment focus that holds both patient and employer accountable, and by exercising appropriate control over the duration of treatment before an MMI determination is made (discussed later) to help ensure optimal results for both the patient and the system. The physician may choose to participate only in the selected phase, such as impairment and disability evaluation, which requires an understanding of the process of an IME or evaluation, as well as application of the appropriate impairment rating guidelines that may vary by jurisdiction.

Compensation and Pensioning Under the Veterans Benefits Administration

In 1930, President Herbert Hoover signed an executive order to establish the US Veterans Administration. The Veterans Administration subsequently underwent multiple expansions and reorganizations over the next 60 years, culminating in the establishment of the US Department of Veterans Affairs (VA) in 1989 as the 14th department within the presidential Cabinet.⁷⁰ The VA's mission is "to fulfill President Lincoln's promise—'*To care for him who shall have borne the battle, and for his widow, and his orphan*'—by serving and honoring the men and women who are America's Veterans."⁷¹ To accomplish this mission, the previous Department of Medicine & Surgery established in 1946 was replaced by the VA's Veterans Health Administration (VHA), which is now the largest integrated health care system in the United States, with over 9 million veterans enrolled.⁷⁸ In addition, the previous Department of Veterans Benefits established in 1953 was replaced by the VBA, which provides a wide range of benefits to service members, veterans, and their families, including disability compensation.⁷⁷

VA disability compensation "is a tax free monetary benefit paid to Veterans with disabilities that are the result of a disease or injury incurred or aggravated during active military service."^{50,74} Eligibility for VA disability compensation requires that the veteran must have been "discharged from service under other than dishonorable conditions," and that the "disability must be the result of an injury or disease that was incurred or aggravated while on active duty or active duty for training; or from injury, heart attack, or stroke that occurred during inactive duty training," meaning that the disability is *service connected*.⁷⁴ Under certain circumstances, disability compensation may also be granted for secondary service-connected disabilities, which develop after completing military service if the disabilities are determined to be "related or secondary to disabilities occurring in service," and for presumptive service-connected disabilities "presumed to be related to circumstances of military service, even though they may arise after service."^{50,74} Most recently, the PACT Act² was signed into law in 2022, expanding "benefits for Veterans exposed to burn pits, Agent Orange, and other toxic substances."⁷⁵

To apply for VA disability compensation, a veteran completes a standardized application, which includes a combination of active-duty medical records, VA medical records, private medical records, and/or supporting statements from friends, family, and fellow veterans who served with the applicant. This package is reviewed by the Adjudication Division of the Compensation and Pension (C&P) Service within the VBA, which usually involves medical examination. VA C&P medical examinations may be performed by physicians, nurse practitioners, physician assistants, psychologists, optometrists, audiologists, and "other qualified" clinical personnel. The VHA oversees C&P examiners and ensures that they are adequately qualified, and all C&P examination reports must be signed by a physician or psychologist. Using a standardized Disability Benefits Questionnaire (DBQ) for the diagnosis in question,⁷² the clinical examiner is asked to render an opinion as to diagnosis of the ratable condition, to address permanency of the condition, and to opine as to whether the individual with the condition is considered totally disabled (fails to meet minimal employability criteria), which is defined as physical inability to be employable at a sedentary level, or psychiatric or psychological inability to be employed in a loosely supervised situation with minimal exposure to the public.⁵⁰ Using the

resulting DBQ and all other supporting materials, a VBA C&P Adjudication Division representative renders a decision if the applicants disability is service connected or nonservice connected based on defined criteria⁹¹ and assigns a rating percentage for resulting service-connected disabilities according to the VA's Schedule for Rating Disabilities.^{79,92} When a veteran has more than one service-connected disability, an overall combined service-connected rating is calculated using the "whole person theory" method⁷⁶ (see later).

Disability compensation is paid as a monthly stipend to veterans based on the combined VA disability rating. The amount of monthly compensation received is based on the current veterans disability compensation rates.⁷³ Disability compensation is not subject to federal or state income tax, it varies according to number of dependents, and it is regularly adjusted to reflect changes in cost of living. Benefits may also include disability pensions for veterans of low income according to a means test, who are permanently and totally disabled, and who have experienced 90 days or more of active duty, at least 1 day of which was during wartime. Other benefits include insurance benefits, specially adapted housing, motor vehicle modifications, and durable medical equipment.^{50,74}

Personal Injury Claims

American law recognizes physical or psychological injury as a personal injury for which monetary damages can be awarded under the law of torts.^{20,40} A claim is usually made against the defendant for personal injury arising out of negligence and in some instances an intentional act. Common examples include those claims arising out of motor vehicle accidents, slip and fall claims against property owners (both private and business), defective products, medical negligence, hospital and nursing home negligence, assault claims, and work-related injuries outside of the aegis of workers' compensation. Plaintiffs are entitled to monetary awards for both actual and general damages (e.g., pain and suffering, nonpecuniary damages). The severity of the injury is the main driver for the compensation, and physician-independent medical examiners (as expert witnesses) are retained by both sides to evaluate the plaintiff's claims of personal injury. The physician typically is required to evaluate the causation, the nature, and the severity of the injury; the exacerbation or aggravation of the pre-existing pathologic condition, if any; and apportionment. In some instances, the physician may also be called on to comment on the necessity of the treatment previously provided or future treatment proposed for the condition under question.

Impairment Rating Guides for Physicians, With Attention to AMA Guides, Sixth Edition

As described in the previous sections, there is a high degree of variability among and within the various US disability systems and significant nuances that distinguish them from one another. Nevertheless, the process whereby a disability determination is made uniformly requires an initial medical determination of an impairment rating based on the schedule recognized by the particular jurisdiction and according to specific medical criteria. Historically, the physician is charged and empowered to render impairment ratings. In response, the American Medical Association (AMA) has produced a rating manual to assist physicians in measuring and rating medical impairments according to a commonly accepted method and scale for assigning impairment

percentages to the “whole person” unit. The *AMA Guides to the Evaluation of Permanent Impairment* (or *AMA Guides*)⁶¹ is a standardized and objective reference and reporting guide for physicians, other health care providers, and other professional stakeholders, including attorneys, adjudicators, insurance adjusters, and the like. The *AMA Guides* was originally published in 1971 and underwent periodical print revisions and updates until the sixth edition published in 2008.⁶¹ Subsequent revisions and updates to the *AMA Guides* have been published and maintained online by the AMA as a “dynamic, digital platform,” which “updates and reinforces the international standard for the assessment of permanent impairment first published in 2008” to provide “access to the most current impairment rating information online.”¹¹ Each revision incorporates the latest scientific evidence and input from a panel of contributors who are recognized experts in their respective medical or surgical fields and represents a consensus view developed through a modified Delphi process to reflect the prevailing medical opinions on organ system impairment ratings. The *AMA Guides* has gained widespread recognition both nationally and globally. For most jurisdictions within the US state and federal workers’ compensation systems, various editions are either mandated or recommended by statute as the standard reference for impairment ratings. It was adopted by the US Department of Labor in 2009 for use with the Federal Employees’ Compensation Program,⁸⁸ and it is the preferred reference for many domestic personal injury claims. Further, it is gaining widespread international adoption as the preferred reference for disability claims in Canada, The Netherlands, South Africa, Australia, New Zealand, Korea, and Hong Kong.^{54,60}

The sixth edition of the *AMA Guides* has adopted the ICF terminology, definitions, and conceptual framework of disablement discussed earlier.^{11,108} They define impairment rating as a “consensus-derived percentage estimate of loss of activity reflecting severity of a given health condition and the degree of associated limitations in terms of activities of daily living.”¹¹ As such, they have adopted metrics sensitive and specific to medical (i.e., anatomic and physiologic) aspects of organ system disease and functional aspects (mobility and self-care) of losses that can occur. These subject areas are considered well within the sphere of expertise expected for physicians who typically evaluate and treat patients with disabilities, including psychiatrists. At the same time, they have avoided metrics that focus on the participation restrictions domain of disablement (e.g., “burden of treatment compliance,”²³ losses in terms of advanced or “instrumental” activities of daily living [ADL], and quality of life⁴⁷) because measuring these requires a level of evaluative skills and training typically outside of the sphere of knowledge of most physicians.⁶⁰

The sixth edition also maintains an important focus on, and inclusion of, four essential elements of physician evaluation and reporting about patients, as follows:

- What is the clinical diagnosis?
- What difficulty does the patient report (symptoms, functional loss)?
- What are the examination findings?
- What are the results of clinical studies?

To better address functional aspects of the impairment rating, the sixth edition has adopted an ADL-based functional history and ordinal measures of ADL assessment to serve as important modifiers of the final impairment rating where applicable.¹¹ This is particularly important to any impairment rating of the musculoskeletal system (discussed later).

Despite the common assumption that physicians by their training are skilled in evaluating impairment or disability resulting from work-related injury or illness, the reality is that physicians in general never receive any formal training in impairment or disability assessment during their medical education or post-doctoral residency training.^{43,52} In response, a number of continuing medical educational venues now exist to provide ongoing training in the use of the impairment rating guides and other aspects of disability medicine. These can be readily accessed through the following organizations and entities: the AMA,¹⁰ the American Academy of Orthopedic Surgeons,⁴ the International Academy of Independent Medical Evaluators,³⁴ the American Board of Independent Medical Examiners,⁶ and the American College of Occupational and Environmental Medicine.⁷

Application of the *AMA Guides*, Sixth Edition, to Musculoskeletal Impairments

The following brief overview is intended to orient the reader conceptually to the musculoskeletal section of the *AMA Guides*, Sixth Edition. The reader is encouraged to consult the *AMA Guides* directly for greater detail and discussion than space permits here.^{11,61}

Qualitative Impairments

Qualitative impairments are anatomically based and belong to discrete, mutually exclusive categories that can only be measured in descriptive terms. Nominal or ordinal scales of measurement can apply to such groupings to yield a hierarchic assembly, but the actual magnitude of such difference between groupings is nonuniform and lacks true proportionality to assigned numeric value. Examples of qualitative impairments recognized for the musculoskeletal system include amputation, joint ankylosis, sensory changes (present compared with absent), and disfigurement or deformity (present compared with absent).

Quantitative Impairments

Quantitative impairments are also anatomically based. They are measured according to continuous scales (interval or ratio) whose units represent fixed values, the ordering of which reflects a uniform and consistent increase in magnitude. The *AMA Guides* recognizes loss of motion (in degrees) in each cardinal plane of function for a given joint as representing quantitative impairment relative to the normally accepted range of motion (ROM) for that joint. Although ROM has intuitive appeal as a standardized, objective, and quantitative measure of joint function, its practical application remains problematic and has been subject to much criticism for its lack of reliability, reproducibility, validity, and feasibility in a typical impairment rating clinical setting.^{11,60,61} Consequently, ROM has been abandoned as a separate rating criterion for the spine; however, ROM still serves as an impairment class (IC) differentiator for some “diagnosis-based impairments” (DBIs; see later) of the upper and lower extremities, and as a grade modifier (GM) within ICs for others; it has also been retained as part of a stand-alone rating for joints of the upper and lower extremities when the DBI method cannot be readily applied (e.g., burn injuries with multiple contractures) or does not fully capture the impairment (e.g., combining proximal losses in joint ROM in the presence of more distal limb amputations).

Diagnosis-Based Impairment Method

The AMA *Guides*, Sixth Edition, has adopted a DBI method as an outgrowth of an earlier “diagnosis-related estimate” approach; there are important additions and changes that are designed to overcome weaknesses of earlier approaches and that are most evident in the musculoskeletal organ system. For example, a uniform platform has been adopted that applies a template (grid) with five columns for mutually exclusive functionally based ICs of incremental severity (classes 0–4) patterned after the ICF classification scheme. Whereas all accepted diagnoses for musculoskeletal conditions affecting the spine or extremities can potentially be viewed within such an ordinal hierarchy, not all conditions will qualify for the higher-class ratings. Accordingly, accepted conditions are hierarchically arranged as a succession or rows beginning with the least severe ratable conditions (by convention, typically soft tissue injuries, sprains, and strains) at the top and ending with the more severe conditions (muscle and tendon traumas, followed by ligament, bone, and joint conditions) at the bottom.^{60,63}

To use this rating system properly, the rating physician must initially determine an accepted diagnosis for the condition being rated and establish permanency in terms of MMI (see later). The first step of the DBI method is to select the most appropriate IC within the most appropriate DBI grid. For the musculoskeletal system, the DBI grids are regionally defined. The spine is divided into four respective regional grids: cervical spine, thoracic spine, lumbar spine, and pelvis. The upper extremity (limb) is also divided into four respective regional grids: digit, wrist, elbow, and shoulder; the lower extremity (limb) is divided into three respective regional grids: foot and ankle, knee, and hip. Once the diagnosis has been established and linked to the most appropriate regional DBI grid, the IC assignment can be made. Each IC will have an available range of five discrete impairment rating scores, with the middle score being the default value for the class. The second step follows, whereby three separate GMs are independently used to determine the level of severity within the IC (i.e., grade on a scale of 0–4) according to functional history (GMFH), physical examination findings (GMPE), and clinical study results (GMCS), respectively. A final step is to calculate the sum of the differences in numeric severity of the impairment grade modifiers minus the IC numeric and summate. It is then possible to triangulate the final impairment score within the IC according to the formula: $(GMFH - IC) + (GMPE - IC) + (GMCS - IC)$. If the sum is zero, then the final impairment rating remains at the middle (default) value; if the sum is +1 or –1, then the impairment rating moves one position to the right or left, respectively; if it is +2 or –2, then it moves two positions to the right or left, respectively.

The DBI method has multiple advantages over earlier approaches. It enables the rater to capture important and useful information on clinical severity and functional disability for a given individual and ratable condition and to weigh the same according to precise criteria of severity rather than solely relying upon clinical judgment. It plays to the strength of the physician rater as a diagnostician and offers greater precision and resolution for a broad array of diagnostic choices than was previously possible. Finally, it offers methodologic uniformity, transparency, and ease of application within and between the organ systems of interest in all cases.^{59,60}

Spine and Extremities as Regional Units

The spine is divided into four discrete regional anatomic units: the cervical, thoracic, and lumbar spine, and the pelvis. Qualitative

impairments include DBI listings as separate “grids” for each unit. Quantitative impairments of the spine do not apply because the ROM method has been removed for this section of the AMA *Guides*.^{11,61}

The upper extremity* is divided into four discrete regional anatomic units: the digits and hand, wrist, elbow, and shoulder. Qualitative impairments include DBI listings as separate grids for each unit as well as separate sections for peripheral nerve injuries, including causalgia (i.e., complex regional pain syndrome type 2 [CRPS2]), CRPS1, and amputations. Quantitative impairments include ROM as a stand-alone. Ankylosis is a qualitative modification to the ROM tables, which are otherwise organized into discrete ranges to which specific impairment numbers are assigned.

The lower extremity* is divided into three discrete regional anatomic units: the ankle and foot, knee, and hip. Like the upper limb, qualitative impairments include DBI listings as separate grids for each unit as well as separate sections for peripheral nerve injuries, including causalgia (i.e., CRPS2), CRPS1, and amputations. Quantitative impairments include ROM as a stand-alone. Ankylosis is a qualitative modification to the ROM tables, which are otherwise organized into discrete ranges to which specific impairment numbers are assigned.

Combining Impairment to Whole Person Impairment Ratings

When dealing with multiple impairments in a single claim, the AMA *Guides* provide a method for combining impairment rather than adding to avoid 100% plus final number. Combining is accomplished with the combined values chart provided in the appendix of the AMA *Guides* and is based on the principle that if two or more values are to be combined, the largest one comes out of the 100% first and the subsequent values come out of the remaining percentage of the “whole person.” The values are derived from the formula: $A + B(1 - A) = \text{combined value of A and B}$, where A and B are the decimal equivalents of the impairment ratings to the same anatomic unit.

The general rule is that all impairments are combined with few exceptions (see later). Combining must occur at the same hierarchal level; for example, upper extremity impairment can only be combined with another upper extremity impairment from the same limb, and whole person impairment can only be combined with another whole person impairment. In the case of impairments from different limbs (either from both upper or lower limbs), even though expressed at the same hierarchal limb level, they should only be combined at the whole person impairment level after each individual limb is fully rated, and the final impairments for each limb are expressed at whole person impairment level.

Exceptions to the “combine rule” include⁵⁵:

- Impairments from ROM loss at the same joint are always added.
- All hand impairments for multiple digits as expressed at hand unit level are added.
- Impairments from ROM loss from all thumb joints are added.
- Palmar digital sensory losses from radial and ulnar digital sides are added.

*Note that *limb* and *extremity* are used interchangeably in the AMA *Guides*, Sixth Edition; *limb* is technically the anatomically correct term because the extremity is the terminal portion of the limb.

Independent Medical Examination: Elements and Reporting Expectations

An IME is (typically) a one-time evaluation performed by a physician who is not directly involved in treating the claimant for the express purpose of answering a series of interrogatory questions posed by the referring party to proceed toward resolution of medical ambiguities and achieve claim settlement. The information gathered by an IME can be used as evidence in litigation claims and other legal proceedings. It may help to untangle the often complex interrelationship between a pathologic condition that may be attributable to a compensable injury and resulting impairment, activity limitations, and participation restrictions in a given case. The opinions set forth by the independent medical examiner must be expressed in terms of medical possibility versus probability according to the definitions and conventions described in the following sections.

Diagnosis and Medical Necessity

The physician examiner is expected to provide the specific diagnosis for each and all allowed (and potentially relatable) conditions relevant to a specific claim. They must also address the medical necessity and appropriateness of diagnostic testing and medical, surgical, and rehabilitative treatment undertaken to date in relation to each of the relevant diagnoses.

Causation and Allowable Conditions

Of the various key questions asked of a physician when serving as an IME or expert witness is the issue of causation. This is an important question because of its significant economic implications for the parties involved. The term *causation* may have different contextual meanings in law compared with medicine, and the physician needs to understand this difference.⁴⁶

Medical causation³¹ is biologic in nature and is established through scientific analysis of sufficient rigor to demonstrate a cause-and-effect relationship with a high degree of certainty (e.g., with a statistical probability [or p-value] of $\leq .05$ [or the probability of being wrong $\leq 5\%$]). For example, a physician can reasonably conclude, within medical probability, that asbestos can cause mesothelioma in an individual exposed to asbestos based on a review of the credible medical evidence in the peer-reviewed scientific literature that has established a causal relationship in this case.

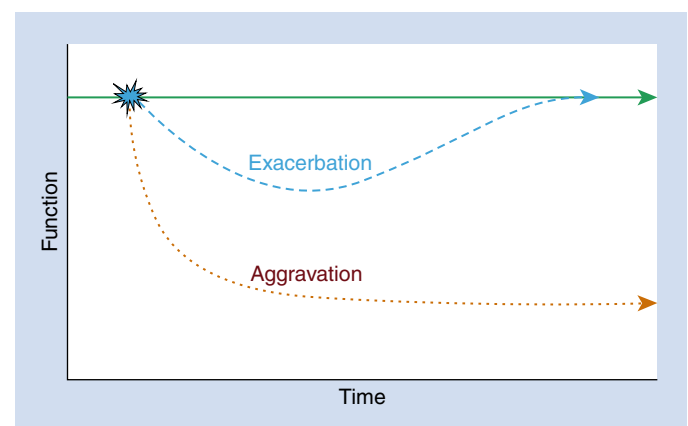
Legal causation,¹⁹ as defined in civil litigation, generally has two prongs. First, an act (e.g., a tort^{20,40}) must be the cause in fact of a particular injury, which means that an act or omission was a necessary antecedent to the personal injury. Legally, this issue is analyzed by determining whether the injury would have occurred “but for” the act alleged to be the cause. If an injury would have occurred independent of the alleged act or omission, cause in fact has not been established, and no tort has been committed. When multiple acts/factors have led to a particular injury, the alleged act or omission is determined to be the cause in fact only if the evidence demonstrates this to have played a substantial role in causing the injury. Second, it must also be established simultaneously that the alleged act was the proximate cause of an injury before the legal liability will be imposed. The concept of proximate cause is very critical because it limits the scope of liability to those injuries that bear some reasonable relationship to the risk created by the wrongdoer. Proximate cause is evaluated in terms of whether

a reasonable person should have foreseen the injury resulting from the act.¹⁸ If a given risk could not have been reasonably anticipated, proximate cause has not been established, and no liability will be attached.

In summary, legal causation is mainly a question of foreseeability.¹⁸ An actor is liable for the foreseeable but not the unforeseeable consequences of an act. For example, it is foreseeable that someone who is left alone on a beach in a drunken stupor at low tide may die from drowning in the rising tide rather than from the excessive alcohol or drugs ingested. However, it is not foreseeable that such an individual will be struck by lightning and killed by that event. In such case, the liability for drowning could have a proximate cause in law (for causing death) for anyone who is found responsible for contributing directly to someone’s drunken stupor and leaving that person exposed to risk of death by drowning, but not to risk of death from being struck by lightning because of its remoteness in probability.

Two important concepts and terms pertaining to causation in relation to a preexisting and underlying condition are *aggravation* and *exacerbation*. Aggravation refers to a permanent worsening of a preexisting condition that occurs when a physical, chemical, biologic, or other factor results in an increase in symptoms, signs, or impairment that never returns to baseline, or what it would have been except for the aggravation. Exacerbation refers to a temporary worsening of a preexisting condition after which the individual recovers to a baseline functional status, or what it would have been had the exacerbation never occurred. The differences between these two contingencies are illustrated in Fig. 6.5.

The workers’ compensation systems vary across states in terms of their causation and work-relatedness rules.⁸⁶ Physicians must understand their state-specific rules and be proficient in following the same to be successful. In some workers’ compensation claims, certain conditions may have been administratively approved to have been caused by the workplace. These include “allowed conditions” in the claim. Certain legislation creates special presumptions for causation for certain workgroups establishing rights for workers and liabilities for employers even in the absence of scientific evidence (and thus medical causation). For instance, many states have laws establishing presumption that cardiovascular diseases, such as hypertension and myocardial infarction, are work related if they occur in firefighters or law enforcement workers as a result of the presumption of the stressful work environment causing these conditions. A law enforcement officer who may



• **Fig. 6.5** Exacerbation versus aggravation of a preexisting condition, in relation to baseline function (green horizontal line).

have worked a lifetime in administration and may have never actually been in the highly stressful situations of the job may still find that a naturally occurring cardiovascular disease, as adjudicated by the workers' compensation system, is considered to be work related. Physicians confronted with these realities of the modern workers' compensation legislation need not be frustrated because these are societal decisions for distribution and redistribution of resources based on prevailing social values and political influences at each jurisdictional level.

Maximum Medical Improvement Determination

The physician examiner is typically required to complete an initial assessment and report, supplemental interim reports, and a final statement at the time of case disposition addressing MMI.⁶⁴ Reporting requirements vary by state, but they are generally similar and must include an estimate of when MMI occurred or is expected to occur. The decision of MMI must be rendered before case closure can be achieved. From a rehabilitative perspective, a claimant should not be considered at MMI as long as expectations for continuing functional improvement are being met by demonstrable and ongoing performance gains. The distinction between functional and pain-oriented goal setting must be maintained during treatment so that progress toward goal achievement can be measured and monitored objectively. When functional progress is no longer evident or tenable and a sufficient (typically 6-month) healing period has transpired, MMI is generally thought to have occurred.⁶⁴

The physician examiner is also required to determine the exact date of MMI and to address issues of medical stability from that point forward. Deterioration that might normally be expected with the passage of time (e.g., progression of an osteoarthritic condition) does not preclude MMI determination. The physician should also address issues of future medical management and follow-up anticipated to be necessary to maintain MMI for a given condition.⁶⁴

Disability as Return-to-Work Restrictions

During the initial evaluation, interim treatment, and MMI phases of reporting, the treating physician or disability examiner will be asked to complete several standardized forms specific to the employer or jurisdiction (federal⁸⁷ or state⁸⁶), including status report and return-to-work/fitness-for-duty forms. As an example, all US Department of Labor OWCP forms are available online⁸⁴; the first page of the *Duty Status Report* (Form CA-17) is provided in Fig. 6.6.

If treatment is ongoing and transitional work is available, the physician might recommend modified duty in terms of restrictions on the number of hours of work allowed and the permissible activities in terms of frequency or degree of material handling tolerated during the healing period. If no modified duty options are available or apply, the physician may be required to render a temporary total disability determination until MMI is reached. Because the probability of returning to work decreases precipitously as time out of work increases,^{8,104} the physician should make every effort to return the claimant safely to a transitional work setting as soon as possible. In cases where transitional return-to-work options are not available, work conditioning and work hardening may be a preferable and viable alternative to forced inactivity and should be considered where feasible and medically necessary.⁶⁴ At the point of MMI, the physician must

also render a final opinion on permanent restrictions applicable going forward.

A number of standardized assessment tools are available to assist physicians in determining valid and reliable physical performance expectations for an injured worker. The functional capacity evaluation⁶⁸ is a comprehensive and standardized therapy assessment of an injured individual's strength, flexibility, endurance, and safety in performing job-specific activities and related material-handling functions. It can provide potentially useful functional metrics to help establish a performance baseline and treatment goals for the injured worker, to monitor recovery as demonstrated by rate of progress toward achieving functional goals during treatment, and to establish a new performance baseline when treatment has been completed for purposes of recommending return-to-work restrictions and accommodation, if needed.⁶⁴ The impaired individual's performance during testing can be assessed in terms of degree of effort, consistency, and reliability. In cases where there is a demonstrated lack of consistency or incomplete effort on the part of a claimant, the validity of test results remains suspect and therefore must be interpreted with caution if not dismissed outright.⁶⁴ As mentioned earlier, a job description is frequently available from the employer and can provide a useful list of the essential functions of the job in question for purposes of identifying required material-handling capacities and addressing specific restrictions that may apply. A job site evaluation can be carried out by a specially trained therapist to validate the essential functions listed in the job description with respect to critical physical demands and relative amounts of time spent performing specific activities within each function. In some cases, ergonomic analysis can be useful to help quantify the job physical demands and enable accommodation in terms of job redesign or workplace modification. In addition, employer and claimant willingness and ability to comply with recommended accommodations can also be addressed.

Physician examiners should avail themselves of these assessment tools to ensure that the prognostic inferences derived, and sanctions imposed, are founded on valid, empiric, and functionally based data, to the fullest extent possible.

Future Medical Needs

While the scope and content of life care planning according to future medical needs is an extensive topic, it is a burgeoning field with which all physiatrists should be familiar. Physiatrists are uniquely qualified to opine upon future aspects of medical care needs in the injured population to maintain MMI and are increasingly becoming interested in this nuanced aspect of medicolegal work. One definition of a life care plan is "a dynamic document based on published standards of practice, comprehensive assessment, data analysis, and research that provides an organized concise plan for current and future needs with associated costs for individuals who have experienced catastrophic injury or have chronic health care needs."³⁶ Life care planning offers a flexible and relatively low-risk way for physiatrists to engage in medicolegal work.

The sequelae of injuries involve several aspects of medical care over the course of a person's lifetime. As detailed in Box 6.1, aspects of medical care might include, but are not limited to, scheduled medical/physician visits, medication prescriptions, invasive and noninvasive medical procedures, serology/laboratory investigations, therapy assessments (such as

Duty Status Report

Reset

Print

U.S. Department of Labor

Office of Workers' Compensation Programs



This form is provided for the purpose of obtaining a duty status report for the employee named below. This request does not constitute authorization for payment of medical expense by the Department of Labor, nor does it invalidate any previous authorization issued in this case. This request for information is authorized by law (5 USC 8101 et seq.) and is required to obtain or retain a benefit. Information collected will be handled and stored in compliance with the Freedom of Information Act, the Privacy Act of 1974 and the OMB Cir. A-130. Persons are not required to respond to this collection of information unless it displays a currently valid OMB control number.

OMB No. 1240-0046

Expires: 08/31/2026

OWCP File Number
(If known)**SIDE A - Supervisor:** Complete this side and refer to physician**SIDE B - Physician:** Complete this side

1. Employee's Name (Last, first, middle)

8. Does the History of Injury Given to You by the Employee
Correspond to that Shown in Item 5? ☐ Yes ☐ No (If not, describe)

2. Date of Injury (Month, day, yr.)

3. Social Security Number

4. Occupation

9. Description of Clinical Findings

5. Describe How the Injury Occurred and State Parts of the Body Affected

10. Diagnosis(es) Due to Injury

11. Other Disabling Conditions

6. The Employee Works

Hours Per Day

Days Per Week

12. Employee Advised to Resume Work?

☐ Yes, Date Advised☐ No7. Specify the Usual Work Requirements of the Employee. Check
Whether Employee Performs These Tasks or is Exposed Continuously
or Intermittently, and Give Number of Hours.

13. Employee Able to Perform Regular Work Described on Side A?

☐ Yes, If so☐ Full-Time or☐ Part-Time

____ Hrs Per Day

☐ No, If not, complete below:

Activity	Continuous	Intermittent		Continuous	Intermittent	
a. Lifting/Carrying: State Max Wt.	#lbs.	#lbs.	Hrs Per Day	#lbs.	#lbs.	Hrs Per Day
b. Sitting	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
c. Standing	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
d. Walking	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
e. Climbing	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
f. Kneeling	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
g. Bending/Stooping	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
h. Twisting	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
i. Pulling/Pushing	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
j. Simple Grasping	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
k. Fine Manipulation (includes keyboarding)	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
l. Reaching above Shoulder	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
m. Driving a Vehicle (Specify)	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
n. Operating Machinery (Specify)	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
o. Temp. Extremes	☐	☐	____ range in degrees F	☐	☐	____ range in degrees F
p. High Humidity	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
q. Chemicals, Solvents, etc. (Identify)	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
r. Fumes/Dust (Identify)	☐	☐	Hrs Per Day	☐	☐	Hrs Per Day
s. Noise (Give dBA)	☐	☐	dBA Hrs Per Day	☐	☐	dBA Hrs Per Day

t. Other (Describe)

14. Are Interpersonal Relations Affected Because of a Neuropsychiatric Condition?
(e.g. Ability to Give or Take Supervision, Meet Deadlines, etc.)☐ Yes ☐ No (Describe)

If you have a disability and are in need of communication assistance (such as alternate formats or sign language interpretation), accommodations and/or modifications, please contact OWCP. See form instructions for Requests for Accommodations or Auxiliary Aids and Services

15. Date of Examination

16. Date of Next Appointment

17. Specialty

18. Tax Identification Number

19. Physician's Signature

20. Date

CA-17 (Rev. 04/2020)

• **Fig. 6.6** US Department of Labor Duty Status Report (Form CA-17). (Courtesy US Department of Labor Office of Workers' Compensation Programs. <https://www.dol.gov/agencies/owcp/FECA/regs/compliance/forms>.)

• BOX 6.1 Categories That May Be Included in a Typical Life Care Plan

Projected evaluations
 Projected therapeutic modalities
 Diagnostic testing, inclusive of education assessment(s)
 Wheelchair and mobility needs
 Mobility accessories and maintenance
 Assistive technology for independent functioning
 Orthotics and/or prosthetics
 Home furnishings and accessories
 Medications and supply needs
 Home care and/or facility care needs
 Future routine medical care needs
 Transportation
 Health and strength maintenance
 Architectural renovations
 Potential complications
 Future medical care and/or surgical interventions, including aggressive treatments
 Orthopedic equipment needs
 Vocational and educational plan(s)

Data from Johnson CB, Weed RO. The life care planning process. *Phys Med Rehabil Clin N Am*. 2013;24(3):403-417.

physical, occupational, speech, and psychological support), and durable medical equipment needs/repair. Quantitative assessment of lost income and absence(s) from the workplace environment are also important pieces of information. A life care plan takes into consideration the duration of a person's expected life, quantifies future needs, and "regularly and reliably employed methods used to formulate the costs" in the plan.³⁶ Although the nuances of life care planning can be explored in other texts, the core process of a typical life care plan remains relatively uniform.

As is explained in "The Life Care Planning Process,"³⁶ a life care plan commences with a referral and case information. This is often followed by acquisition and organization of documentation to support the case, including comprehensive review of medical records.^{16,36} A face-to-face clinical interview of the client is incorporated into the plan. A life care planner may prepare a preliminary life care plan from this information, but this plan often requires additional information, which is substantiated from supplemental consultations and evaluations. The life care planner may research costs and sources of information before finalizing the plan.^{16,36}

Life care planning is an emerging field; as such, it encompasses a variety of professionals who may act in the role of a life care planner. Currently, there are no stipulations that require the life care planner to be a physician and no single governing body that standardizes life care planning certification. There are several avenues of certification, which are beyond the scope of this introductory material. Because of the complex nature of this field, however, it is generally recommended to receive some additional training in life care planning. There are medicolegal implications that could adversely impact the life care planner if poorly completed, and care should therefore be taken to best prepare a thorough plan. For more information, the reader is directed to available textbooks,^{16,65} journal series⁴¹ (which includes specific articles on life care planning for individuals with amputation,⁴⁵ spinal cord injury,⁶⁹ and traumatic brain injury¹¹⁰), and online resources from the American Academy of Physician Life Care Planners.⁵

Legal and Ethical Considerations

Expert Witness Testimony

Physiatrists appear uniquely qualified among medical specialists to perform impairment ratings and disability evaluations because the scientific and medical foundation essential to physiatry maintains a focus on human functioning. The knowledge, skills, and abilities of any physiatrist can be naturally extended, and the physiatrist can become a routine practitioner within the emerging field of disability medicine, which has been described as a subspecialty of clinical medicine encompassing the identification, prediction, prevention, assessment, evaluation, and management of impairment and disability, both in individuals and across populations. Thus far, a major thrust of this field has been the physician's role in providing expert witness testimony regarding personal injury claims within a medicolegal context.⁵³

The underlying principle of common law¹⁷ (both law of tort and contracts) is to provide a venue for a person who has suffered damage as a result of an action or inaction by others to seek redress for personal grievance in the courtroom.⁵³ Tort law has been described as a great equalizer because it gives the individual seeking redress the ability to bring any offending employer or company (referred to in legal parlance as a *tortfeasor*) before the bar on a more equal footing for alleged wrongdoings. Obviously, the law cannot make tortfeasors undo their injury or harm, but it does make them accountable through monetary compensation for both intentional wrongs (e.g., defamation, assault, and battery) and unintentional wrongs (e.g., negligence). The law of tort not only shifts the burden of the cost of the injury or damages to the responsible party but serves to prevent similar harm to other members of society through enforced accountability. The idea is to make the offensive and undesirable behavior costly to the tortfeasor and, in principle, deter others from engaging in such future offenses.^{20,40,53}

The opinions set forth in an IME and the resulting formal report comprise expert witness testimony that must validate or refute the presence and severity of injury toward, and resulting disability to, any claimant.⁵³ This occurs within a legal infrastructure from which that claimant maintains certain legal rights and entitlements and may derive significant monetary gain. The independent medical examiner, in many instances acting as an agent for the consulting party rather than solely as a patient advocate, typically faces the risk of targeted allegations of wrongdoing leveled by any disgruntled claimant who may view the physician's opinion as adversarial to the claim. Serious allegations of financial,²⁸ physical,^{57,67} or psychological injury²⁹ to patients in this context are a matter of record, and expert witness physicians are being held accountable when those injuries are proven in a court of law.⁵³

Box 6.2 enumerates various legal theories of liability against the IME and expert witness physicians.

Traditionally, courts in all jurisdictions have held that physician expert witness testimony and IMEs are essential to the efficient functioning of workers' compensation and personal injury claim systems and are at once beneficial to employers, insurers, and claimants alike.⁵³ Such testimony provides a clinical foundation for medicolegal decisions that help set the standard for society's definition of health and illness and the legal implications that flow from that standard. As a consequence, the US legal system has tended to grant relative immunity to physicians in such court decisions when acting in an expert witness capacity.

• BOX 6.2 Legal Theories of Causes of Action Against Independent Medical Examiner Physicians and Expert Witnesses

- A. Intentional torts
 - 1. Assault
 - 2. Battery
 - 3. Intentional infliction of emotional distress
 - 4. False imprisonment
 - 5. Defamation
 - 6. Invasion of privacy
 - 7. Fraud and misrepresentation
 - 8. Conspiracy
 - 9. Bad faith
 - 10. Deceptive trade practices
- B. Unintentional torts
 - 1. Ordinary negligence
 - 2. Professional malpractice
 - 3. Failure to warn
 - 4. Wrongful death
 - 5. Loss of chance of recovery or survival
 - 6. Vicarious liability for the acts of others
 - 7. Negligent referral
 - 8. Failure to diagnose
 - 9. Failure to inform
- C. Actions under the law of contract
 - 1. Breach of contract
 - 2. Breach of warranty
 - 3. Abandonment
- D. Miscellaneous causes of action
 - 1. Deceptive trade practices
 - 2. Violation of a statute or regulation

Recent court actions in this area raise concerns that such legal shelter no longer applies.

It is noteworthy that until recently, an expert witness enjoyed essentially the same type of immunity as any other witness in the judicial process. This immunity from civil liability dates back to 16th-century English common law; it was very broad and included protection from claims of defamation and negligence.²² The principal rationale for witness immunity was to ensure that the witness would speak freely when giving testimony.³⁰ The argument for such immunity has been that expert witnesses are an important part of the legal system and that in the interest of justice, expert witnesses also need to be protected from liability. The US courts until very recently have regarded this issue of expert witness immunity to be so important that it was maintained even in cases when there might be negligence.¹⁵

The judicial philosophy of the 19th- and 20th-century American jurisprudence emphasized that the object of immunity is not to protect those whose conduct is open to criticism but to protect those who would be subject to unjustified and vexatious claims by disgruntled litigants. The courts generally have affirmed the concept of expert witness immunity for reasons of public policy because without immunity there would be loss of objectivity. The fear of infinite vexation could also have a chilling effect on any witness, resulting in reluctance to testify.¹³ However, there has been an erosion in expert witness immunity from civil actions during the past 2 decades in some US state jurisdictions. Legal commentators have attributed this, in part, to growth and proliferation of the litigation/expert witness industry and to the courts' perception that there is insufficient protection of the injured party from unscrupulous witnesses and inadequacy of traditional

safeguards against witness malfeasance through perjury and inadequate cross-examination.

Medical malpractice actions against IME physicians as expert witnesses failed until recently because of lack of a defined physician–patient relationship with the examinee or plaintiff.⁵⁶ Within the context of any IME, the contention was maintained that the physician neither offered nor intended nor was authorized to actively treat the individual examinee; hence there was no established physician–patient relationship to sustain the medical malpractice cause of action.^{37,39} Many such cases were either not filed because of a prevailing notion among the legal community based on the previous case law or, when filed, were dismissed on pretrial motions from the defendants.^{21,26,66} This has, however, changed significantly in the past 2 decades, with increasing case law from various jurisdictions holding independent medical examiners and expert witnesses accountable for the alleged harm endured by the plaintiff/examinee. This was initially done under the cause of action of simple negligence and outside the law of torts for medical malpractice.²⁸

However, during the past decade, supreme courts in at least two states have allowed civil action against IME physicians to proceed under the traditional medical malpractice theory. The duty of care to the patient despite the absence of a formal physician–patient relationship is the underlying new cause of action against IME physicians and expert witnesses. The Arizona Supreme Court in *Ritchie v. Krasner*⁵⁷ even allowed medical malpractice to go forward despite no physician–patient relationship by essentially stating that the court can envision no public benefit in not holding a physician accountable to a duty to conform to the legal standard of reasonable care.

In summary, psychiatrists and other practitioners in the field of disability medicine performing IMEs, giving expert testimony, or performing life care planning should be aware of not only the legal liabilities in the overall practice of their subspecialty but also the additional malpractice and civil liabilities entailed with exposure to the practice constraints under which IMEs, expert witness work, and life care planning are performed. It should be emphasized that even though the recent case law in some jurisdictions has significantly removed the traditional malpractice immunity of providers who have no clearly defined physician–patient relationship established with their examinees, considerable demand for expert medical witness services remains. Physicians who are attracted to disability assessment and inclined to serve as independent medical examiners are encouraged to attend several of the high-quality training programs offered in the United States to independent medical examiners, expert witnesses, and life care planners with the goal of empowering them with greater knowledge, skills, and abilities necessary to practice as an independent medical examiner or expert witness in the field of disability medicine.^{5,6,53}

Psychiatrists involved in medicolegal practice can be successful despite these challenges if they remember several key principles, including intellectual honesty, professionalism, and respect for the judicial process at all times.⁵³ An ethical and objective examiner who performs a thorough evaluation, deals with the plaintiff/claimant in an empathetic, unbiased manner, and avoids advocacy has less risk of getting entangled in allegations of wrongdoing.

The physician examiner should be judicious in gathering all the facts and available objective data and base opinions on evidence rather than subjective anecdotal experiences.⁵³ The American judicial process is typically carried out in an adversarial arena where challenges to one's expert opinion are likely to arise,

TABLE 6.1 Guide of Dos and Don'ts for the Physician Providing Expert Witness Testimony

<i>Do</i> – Present a positive image; be respectful of the judicial process and take it seriously	<i>Do not</i> – Lose your temper even though you may be provoked
<i>Do</i> – When appearing in court arrive on time or early, dress conservatively, and be relaxed	<i>Do not</i> – Go beyond your expertise or specialty
<i>Do</i> – Prepare yourself; review all the facts	<i>Do not</i> – Be arrogant or sarcastic
<i>Do</i> – Know both the strengths and weaknesses of your case	<i>Do not</i> – Guess or assume
<i>Do</i> – Always tell the truth	<i>Do not</i> – Compromise your integrity
<i>Do</i> – Listen carefully; understand the question	<i>Do not</i> – Be an advocate
<i>Do</i> – Limit your response to the specific question	<i>Do not</i> – Argue with the opposing counsel
<i>Do</i> – State opinions only when called for	<i>Do not</i> – Answer if you are told not to
<i>Do</i> – Keep it brief by stating a single fact, without going into long needless discussions	<i>Do not</i> – Volunteer unsolicited information or answers
<i>Do</i> – Concede to the opponent's point when reasonable or appropriate	<i>Do not</i> – Discuss the case in public areas

no matter which side is being represented. Opinions based on scientific evidence and sound judgment are therefore much more likely to withstand the adversarial scrutiny and challenge of even the most skilled cross-examiner. Table 6.1 details a list of dos and don'ts that can serve as a guideline for the physician examiner providing expert witness testimony.

Ethical Considerations

The physician examiner must also confront the ethical challenges posed by the patient as a claimant. As has been discussed earlier, there is increasing ambiguity regarding the moral imperative of the traditional physician–patient relationship in these cases. In the arena of workers' compensation, the physician is at once accountable to the client (referral party) whose interests may conflict with those of the patient and the opposing counsel. In addition, the physiatrist, by nature of the interdisciplinary approach, must remain committed to preserving patient autonomy as a member of the treatment team and to maximizing functional recovery (thereby minimizing disability) and reducing or eliminating dependency to the fullest extent possible on the treating system and caregivers, including the disability system itself. The patient, as a claimant, is often represented by legal counsel and may be coached and

instructed or may otherwise choose to behave in a manner that is counterproductive to these goals and thereby appear as noncompliant.

The physician must also remain cognizant of the paradox of compensable injury—that financial compensation can discourage return to work and thereby promote the disability.⁵³ Undue prolongation of an open claim (through inappropriate and excessive diagnostic or therapeutic endeavors, however well intended) might further serve to legitimize disability in the claimant's mind and can inhibit the likelihood of functional recovery and return to work. Decisions to terminate treatment of compensable injuries and to reach MMI might not always be mutually agreeable to claimant and examiner and in most cases are likely to rest on the final authority of the physician rather than that of the patient. Perhaps the most useful beacon to guide decision making when treating compensable injuries is to promote functional recovery to the fullest extent possible, to terminate treatment when functional recovery is no longer tenable, and to render impairment ratings and return-to-work decisions that enable patients to use their residual abilities (through accommodation if necessary) as soon as possible and to the fullest extent possible. (A list of glossary key terms is given in Box 6.3.)

• BOX 6.3 Glossary of Terms

Apportionment: A determination of percentage of impairment directly attributable to preexisting or resulting conditions and directly contributing to the total impairment rating derived.^{11,61}

Disability: An umbrella term for activity limitations or participation restrictions in an individual with a health condition, disorder, or disease.^{1,11,61}

Functional capacity evaluation (FCE): A comprehensive assessment of an individual's strength, flexibility, endurance, and job-specific functional abilities. An FCE includes a feasibility assessment of the individual's ability to perform the essential functions of a specific job and, when validated, may be useful to help predict return-to-work potential and appropriate restrictions in a particular case.^{34,68}

Impairment: A significant deviation, loss, or loss of use of any body structure or function in an individual with a health condition, disorder, or disease.^{11,61}

Impairment rating: A consensus-derived percentage estimate of loss of activity, which reflects severity of impairment for a given health condition, and the degree of associated limitations in terms of activities of daily living.^{11,61}

Job description: A formal listing of the essential functions constituting a particular job in terms of the specific physical performance requirements.^{11,61,84}

Job site evaluation (JSE): An onsite analysis of the workplace to determine optimal ergonomic design and validate specific physical performance requirements of the job. A JSE can be useful in concert with an FCE to determine applicable return-to-work restrictions and to help ensure employer/employee compliance when necessary.^{34,68}

Life care plan: "A dynamic document based on published standards of practice, comprehensive assessment, data analysis, and research that provides an organized concise plan for current and future needs with associated costs for individuals who have experienced catastrophic injury or have chronic health care needs."³⁶

Maximum medical improvement (MMI): The point at which a condition has stabilized and is unlikely to change (improve or worsen) substantially in the next 12 months, either with or without treatment.^{11,53,59,61}

Medical possibility: Something could occur because of a particular cause (probability of occurrence is as likely as not and is ≤50%).^{11,53,61}

• BOX 6.3 Glossary of Terms—cont'd

Medical probability: Something is more likely to occur than not (probability of occurrence is >50%).^{11,53,61}

Permanency: Permanency and MMI are related concepts and simply mean that a person with an injury, after having received adequate medical, surgical, and rehabilitative treatment and achieved clinical and functional stability, is now as good as possible. Other synonymous terms in use according to jurisdictional preference include fixed and stable, maximum medical recovery, maximum medical stability, and medically stationary. In workers' compensation jurisdictions (listed in the text), these terms are useful to enable the injured person to exit the temporary disablement stage of recovery thereby facilitating claim settlement and case closure.^{11,53,61}

Scheduled loss: Allocation of a specified value for purposes of indemnification to a regional anatomic or functional unit to which an impairment rating can

be assigned. The specified value allowed for a given unit can be expressed in terms of weeks or months of lost wages.^{11,55,59,61}

Unscheduled loss: Estimated functional loss to the "whole person" for purposes of indemnification and according to a physiologic system rather than to a regional anatomic or functional unit. The cardiopulmonary, gastrointestinal, and central nervous systems are examples of systems to which an unscheduled loss can apply.^{11,55,59,61}

Work hardening: A work-oriented treatment program delivered in a highly structured environment that simulates the workplace and is designed to improve job productivity of an injured or deconditioned worker. Productivity goals can pertain to work tolerance, job proficiency, or job efficiency.^{34,68}

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